# ***Introduction to Artificial Intelligence***

## Perception and Action Cycle; Types of Environments

### The Perception–Action Cycle

An agent operates in a continuous loop:

Perceive/sense → interpret → decide → act → environment changes → Perceive/sense again.

- Perceive: sensors capture raw data (pixels, signals, text).
- Interpret: the agent updates its understanding of the current situation.
- Decide: it selects an action using rules, a model, goals or utility.
- Act: actuators execute the action.
- Feedback: the changed environment produces new precepts, closing the loop.

Example: a self-driving car sees a red light, decides to stop, brakes, then perceives its reduced speed and the car ahead, and adjusts again.

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### Types of Task Environments

A task environment is simply the situation in which an AI agent works.

We classify environments using different properties.

| Dimension | Meaning | Easier | Harder |
|---|---|---|---|
| **Observability** | Can sensors see the complete relevant state? | Fully observable<br>The agent can access all relevant information about the current state.<br>Example: Chess board where all pieces are visible. | Partially observable<br>Some information is hidden.<br>Example: Poker — you cannot see the other players' cards. |
| **Determinism** | Is the next state fully determined by the current state and action? | Deterministic / Predictable<br>The same state and action produce a predictable next state.<br>Example: A crossword puzzle, assuming the rules and clues are fixed. | Stochastic / Uncertain outcome<br>Outcomes involve randomness or uncertainty.<br>Example: (taxi driving, dice games) |
| **Episodes** | Is each decision independent of past ones? | Episodic<br>Each task is mostly independent.<br>Example: An AI classifies separate images of fruits. One classification does not affect the next. | Sequential<br>Current decisions affect future decisions.<br>Example: (chess, driving) |
| **Dynamics** | Does the environment change while the agent is deciding? | Static<br>The environment remains unchanged while the agent decides.<br>Example: (crossword) | Dynamic<br>The environment may change while the agent decides.<br>Example: (taxi driving) |

| Dimension | Meaning | Easier | Harder |
|---|---|---|---|
| **State/time** | Finite set of states and actions, or continuous? | Discrete<br>A limited or countable set of states or actions.<br>Example: chess have counted moves | Continuous<br>Values can vary over a range.<br>Example: A car's speed, position or steering angle. |
| **No. of Agents** | How many agents act in the environment? | Single-agent (Solving crossword alone) | Multi-agent:<br>Competitive (chess)<br>Cooperative (Multiple robots work together.) |

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## Classifying Some Familiar Tasks

| Task | Observable | Deterministic | Episodic | Static | Discrete | Agents |
|---|---|---|---|---|---|---|
| **Crossword puzzle** | Fully | Deterministic | Sequential | Static | Discrete | Single |
| **Chess with clock** | Fully | Strategic | Sequential | Semi-dynamic | Discrete | Multi |
| **Poker** | Partially | Stochastic | Sequential | Static | Discrete | Multi |
| **Taxi driving** | Partially | Stochastic | Sequential | Dynamic | Continuous | Multi |
| **Image classification** | Fully | Deterministic | Episodic | Static | Discrete | Single |

# Types of Agents

## Simple Reflex Agents

- It looks at the current situation and immediately follows a rule (IF condition THEN action).
- No memory of past precepts, no model of the world.
- Work only when the correct decision can be made from the current percept alone, i.e., a fully observable environment.

```
function SIMPLE-REFLEX-AGENT(percept):
    state <- INTERPRET-INPUT(percept)
    rule  <- RULE-MATCH(state, rules)
    return rule.action
```

Example: a thermostat; an automatic door; a vacuum that sucks if the square is dirty.

Limitation: can loop forever in partially observable settings.

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## Model-Based Reflex Agents

- It uses the current observation plus memory of the world.
- Maintain an internal state: a model of how the world evolves and how actions affect it.
- Use the model to track parts of the world that cannot be seen right now.

Example: Imagine a car changing lanes.

- It sees a vehicle in the side mirror.
- That vehicle temporarily disappears from view.
- The car can still use its internal state to remember that the vehicle may be in its blind spot.

So, it uses:

- Current information
- Previous information
- A model of how the world changes

Main benefit: It can handle partially observable environments better.

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## Goal-Based Agents

- It knows what it wants to achieve and plans how to achieve it.
- Choose actions by considering the future: "Will this action help me reach the goal?
- More flexible than reflex agents: change the goal and the behaviour changes without rewriting rules.

Example: Google Maps-style navigation.

Goal: Reach your college.

The agent considers possible routes and chooses actions that help reach the destination.

If the destination changes, it can plan a different route.

Main benefit: It can think about future results using search and planning.

Current state + Goal → Planning → Action

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### Utility-Based Agents

- A goal-based agent may find multiple ways to reach a destination.
- Used to choose among multiple ways of reaching a goal, to trade off conflicting goals and to handle uncertainty by maximising expected utility.

Example: Route A takes 30 min but cost 100 rupees for toll and Route B takes 40 min but 0 toll

If saving money is more important, Route B may be preferred. If reaching quickly is more important, Route A may be preferred.

Possible outcomes → Utility comparison → Best action

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## Learning Agents

Any of the above can be turned into a learning agent, which improves with experience. It has four conceptual components:

- **Performance element:** Selects and performs actions.
- **Critic:** Checks how well the agent is performing and gives feedback.
- **Learning element:** Uses feedback to improve the agent.
- **Problem generator:** suggests exploratory actions that lead to new, informative experiences.

Example:

**A game-playing AI:**

- **Performance element → Makes a move.**
- **Critic → Evaluates how well it played.**
- **Learning element → Improves its strategy.**
- **Problem generator → Tries new strategies or moves to explore.**

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## Comparison

| Agent type | Uses | Strength | Limitation |
|---|---|---|---|
| **Simple reflex** | Current percept + rules | Fast, simple | Needs full observability; no foresight |
| **Model-based reflex** | Percept history + world model | Handles partial observability | Still no explicit goals |
| **Goal-based** | Model + goals + search | Flexible, purposeful | Cannot rank among goal-achieving options |
| **Utility-based** | Model + utility function | Handles trade-offs and uncertainty | Utility design and computation are hard |
| **Learning** | Any + critic + learner | Improves; works in unknown settings | Needs experience/data; can learn the wrong thing |